

Mid-term examination
Quantum Mechanics & Electromagnetism [PH-1-03(TO)]

Duration: 1.5 hr

Feb 17, 2026

Total Marks: 30

All questions are compulsory. All letters and symbols shown in bold denote vectors. The formula sheet is provided on the back side of the question paper.

1. Show that $\mathbf{E} = k(xy \hat{x} + 2yz \hat{y} + 3xz \hat{z})$ is not an electrostatic field. [3 Marks]
2. A line charge λ is glued onto the rim of a wheel of radius b , which is then suspended horizontally so that it is free to rotate (assume that the spokes are made of some nonconducting material). In the central region, out to radius a , with $a < b$, a uniform magnetic field $\mathbf{B}(t)$ pointing up increases with time. What is the induced electric field in the wheel? [3 Marks]
3. From Maxwell's equations for electrodynamics in a vacuum (see the formula sheet), justify for magnetic ($\mathbf{B}$) and electric ($\mathbf{E}$) fields that $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\partial \mathbf{A} / \partial t - \nabla V$, where $\mathbf{A}$ and V are vector and scalar potentials, respectively. Show that $\mathbf{E}$ and $\mathbf{B}$ remain invariant if you simultaneously transform $\mathbf{A} \rightarrow \mathbf{A} + \nabla \chi$ and $V \rightarrow V + \beta$, where χ is time-independent and β is a constant. [2+2 Marks]
4. A thin metallic wire carrying a steady current $\mathbf{I} = I \hat{z}$ is embedded in a material (permeability μ) which coaxially extends to radius R . Find the auxiliary field $\mathbf{H}$ and the magnetic field $\mathbf{B}$ at every point in the material. Assuming that the material is linear and homogeneous, i.e., the magnetization $\mathbf{M} = \chi_m \mathbf{H}$, find the bound surface current density $\mathbf{K}_b$. [4+2 Marks]
5. The space between the plates of a parallel plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a so that the total distance between the plates is $2a$. The permittivities of slab materials 1 and 2 are ϵ_1 and ϵ_2 respectively. Further, the free charge density on the top and bottom plates are σ and $-\sigma$ respectively. Find the electric displacement $\mathbf{D}$ in each slab. Find the electric field $\mathbf{E}$ in each slab. [4+2 Marks]
6. Using Maxwell's equation in free space, show that the electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields obey 3D wave equation. Write down the uniform plane wave solution for $\mathbf{E}$ and $\mathbf{B}$. Show that electromagnetic waves are transverse waves, i.e., the wave propagates in a direction perpendicular to the directions of $\mathbf{E}$ and $\mathbf{B}$ fields. What is the average power per unit area transported by the electromagnetic fields? Note that the electric and magnetic field amplitudes satisfy $E_0 = cB_0$, where c is the speed of light in vacuum. [2+1+3+2 Marks]
Hint: You may use the identity $\nabla \times (\nabla \times \mathbf{G}) = \nabla(\nabla \cdot \mathbf{G}) - \nabla^2 \mathbf{G}$ to derive the wave equation.

- Maxwell's equation for electrodynamics in a vacuum:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.\end{aligned}\tag{1}$$

- Poynting vector,

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}\tag{2}$$

- Bound volume current density, $\mathbf{J}_b = \nabla \times \mathbf{M}$
- Bound surface current density, $\mathbf{K}_b = \mathbf{M} \times \hat{n}$
- In cartesian coordinates,

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}\tag{3}$$

- In spherical coordinates,

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}\tag{4}$$

Volume element, $d\tau = r^2 \sin \theta dr d\theta d\phi$,

Area element, $da = R^2 \sin \theta d\theta d\phi$

End-semester Examination Quantum Mechanics and Electrodynamics

April 28, 2026

Time Allowed: 3 Hours

Total Marks: 50

1. Given a wave function, $\psi(x) = A\sqrt{x}e^{-x/2}$, $x \in [0, \infty)$.

(a) Find the normalization constant, A . [3 Marks]

(b) Plot the probability distribution function, $p(x)$, over the allowed region. [3 Marks]

2. An electron is confined to a 1D infinite potential well of width 0.5 nm. Calculate the energy (in electron-volts, eV) required to excite the electron from the ground state ($n = 1$) to the second excited state ($n = 3$). [3 Marks]

3. The Schrödinger equation is a linear differential equation in wave function, ψ . A consequence of that fact is: a linear combination of two allowed wave functions for the system is also a valid wave function. Given the following wave function, $\psi(x)$, defined over a region $x \in [0, a]$:

$$\psi(x) = A \left(\sin \frac{\pi x}{a} + \sin \frac{2\pi x}{a} \right)$$

(a) Show whether the wave function is a solution of the Time Independent Schrödinger Equation (TISE). Justify your answer. [3 Marks]

(b) Find the normalization constant, A . [3 Marks]

4. In a 1D finite well, the wave function in the classically forbidden region $x > a$ is given by $\psi(x) = Ce^{-\kappa x}$. If the wave function inside the well ($0 < x < a$) is $\psi(x) = A \sin(kx)$, express the relationship between A and C using the boundary conditions at $x = a$. [3 Marks]

5. A particle in a 1D infinite potential has the following wave function:

$$\psi(x) = \sqrt{\frac{3}{4}}\psi_1(x) + \sqrt{\frac{1}{4}}\psi_2(x)$$

where, $\psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)$, $\psi_2(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)$. Find the total energy of the particle in terms of the ground state energy E_1 . [4 Marks]

6. The final characteristic equation for the idealized Kronig-Penney model is:

$$P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos(ka)$$

Imagine a specific crystal lattice where the scattering power is $P = \pi/4$.

- (a) Calculate the value of the left-hand side of the equation for an electron whose energy E results in the parameter $\alpha a = \pi/2$. Based on your calculation, does this specific energy E lie in an allowed band or a forbidden band (band gap)? Justify your answer mathematically. Note that $\alpha^2 = 2mE/\hbar^2$ and a is the periodicity of the crystal lattice. [3 Marks]

- (b) Explain the limiting conditions, $P = 0$ and $P = \infty$. Justify your answers mathematically. [4 Marks]

7. An inverted hemispherical bowl of radius R carries uniform surface charge density σ . Find the potential difference between the "north pole" and the center. [4 Marks]

8. A crystalline lattice with lattice constant, $2a$, has a periodic potential (see Figure 1). Draw the probability density, $p(x)$, over the entire region for the following electron energies with $k = \pi/4a$.

- (a) $E < V_0$ and with $\psi(x) = e^{ikx} \sin(\pi x/a)$. [3 Marks]
 (b) $E = 2V_0$. [3 Marks]

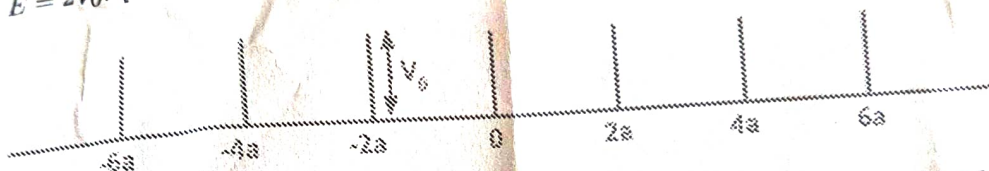
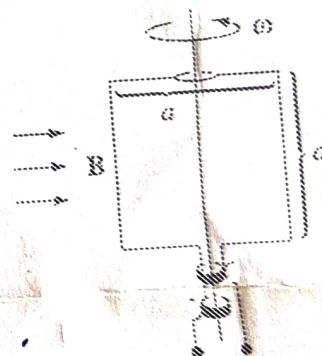


Figure 1: Periodic potential with lattice parameter $2a$ and barrier height V_0

9. Write down the electric and magnetic fields [use sin or cos functions] for a monochromatic plane wave of amplitude E_0 , frequency ω , and phase angle $\phi = 0$ traveling in the direction from the origin to point $(1, 1, 1)$, with polarization in the XZ plane. Sketch the wave and label $\vec{E}$, $\vec{B}$ and $\vec{k}$. [4+2 Marks]



10. A square loop (side a) is mounted on a vertical shaft and rotated at angular velocity ω (see Figure 2). A uniform magnetic field $\vec{B}$ points to the right. Find the emf induced in the loop. [5 Marks]

Figure 2: Rotating square loop

Assumption: In all the QM problems, the systems are conservative, i.e., the energy is constant with time therefore only the spatial part of the wave function, $\psi(x)$ is given.

Physical Constants

- Planck's constant (h): $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$
- Reduced Planck's constant ($\hbar$): $1.054 \times 10^{-34} \text{ J} \cdot \text{s}$
- Mass of electron (m_e): $9.109 \times 10^{-31} \text{ kg}$
- Elementary charge (e): $1.602 \times 10^{-19} \text{ C}$
- $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$